



National Technical University of Athens

School of Electrical and Computer Engineering

8th Semester

VLSI Digital Systems

6th Laboratory Exercise (Bonus)

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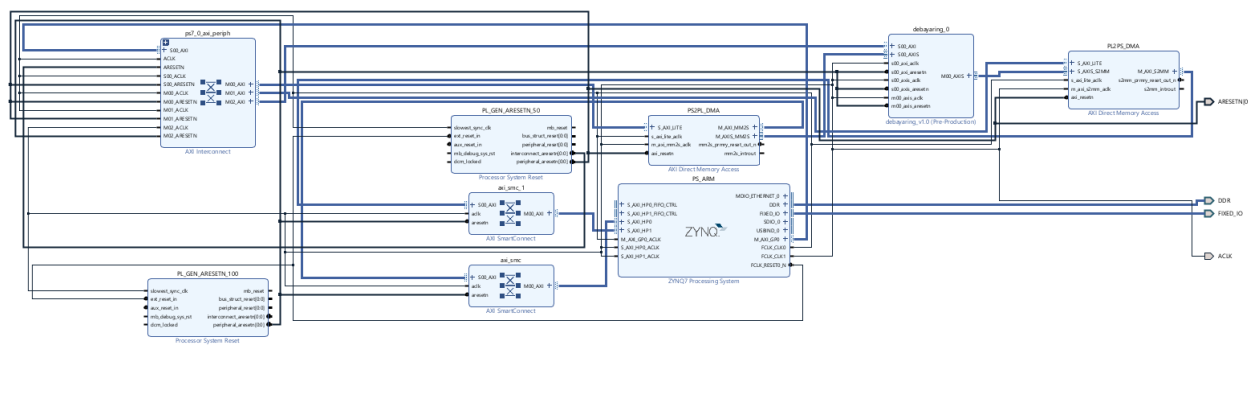
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1 System Block Diagram



2 Brief Explanation of the Main Structural Units of the Circuit

2.1 General Overview

The 6th laboratory exercise extends the 5th exercise, with the main objective of converting the debayering filter into a *runtime reconfigurable* system, i.e., a system capable of dynamically changing the supported image size without reprogramming the FPGA.

The main change compared to the 5th exercise is that the generic parameter N , which statically determined the image dimensions during synthesis, has been replaced by an input signal called `image_dim`. This signal is dynamically configured by the ARM processor through an AXI-Lite interface. In addition, the `image_dim_vld` signal was introduced to indicate the validity of the new value.

Finally, it is worth mentioning that, for the representation of `image_dim`, the second approach proposed in the assignment specification was selected. More specifically, the `image_dim` signal directly contains one dimension of the image. For example, for an image with dimensions 512×512 , we have `image_dim` = 512.

2.2 Serial to Parallel (Runtime Reconfigurable)

The appropriate modifications have been made to the `serial_to_parallel` module in order to support a parametrizable image size at runtime. Specifically, the operations that, in the previous laboratory exercise, were based on the *generic* parameter N have been replaced by corresponding operations using the `image_dim` signal, which is received through the `AXI Lite` interface.

It is worth noting that, since the previous laboratory exercises, the system has already supported the processing of images with different dimensions, provided that their dimensions fit within the FIFO capacity. This was achieved through an appropriately parameterized mapping, which was analyzed in previous reports. Therefore, no significant modification was required in this particular module, as would have been the case if this functionality had not already been implemented.

The rest of the operation remains unchanged compared to Exercise 5. At each clock cycle, a new pixel is inserted into the first FIFO, propagated sequentially through the subsequent FIFOs, and a 3×3 window is formed, which is then used by the calculator.

2.3 Calculator

The calculator remains essentially unchanged compared to the 5th laboratory exercise. It receives as input a 3×3 window of neighboring pixels (`din00` to `din22`), identifies the type of each pixel (GREEN1, GREEN2, BLUE, RED) based on the GBRG Bayer pattern, and calculates the R, G, and B values by applying the appropriate averaging operations.

2.4 FSM (Runtime Reconfigurable)

In this module, as in the other practical modules, no substantial changes to its basic operation were required. The only modification concerns the replacement of the *generic* parameter, which was used in the previous laboratory exercise to determine the image size, with the `image_dim` signal. In this way, the image dimensions can now be determined dynamically at runtime, without requiring the hardware to be resynthesized.

2.5 AXI-Lite Slave Interface

The AXI-Lite Slave interface is responsible for transferring the image dimensions from the ARM processor to the FPGA. The ARM processor writes the desired value of `image_dim` to a register of the AXI-Lite interface, which then generates the `image_dim` and `image_dim_vld` signals for the debayering filter.

2.6 AXI-Stream Slave & Master Interfaces

The AXI-Stream interfaces remain identical to those used in the 5th laboratory exercise:

- **AXI-Stream Slave:** Receives the input pixels from the DMA. The `Slave_TDATA[7:0]` signal is connected to `pixel[7:0]`, while `Slave_TVALID` is connected to `valid_in`.
- **AXI-Stream Master:** Sends the RGB results to the DMA. The `Master_TDATA[31:0]` signal is defined as `"00000000" & R[7:0] & G[7:0] & B[7:0]`, `Master_TVALID = valid_out`, and `Master_TLAST = image_finished`.

2.7 reconfigurable_debayering_filter (Top Module)

The top module of the system integrates all the subsystems described above: the `image_dim` decoder, the FSM, the `serial_to_parallel` converter, the calculator, and the AXI interfaces. It does not contain any additional processing logic, but functions exclusively as an interconnection layer.

The input signals include: `clk`, `rst_n`, `new_image`, `valid_in`, `pixel[7:0]`, `image_dim`, and `image_dim_vld`. The output signals include: `R[7:0]`, `G[7:0]`, `B[7:0]`, `valid_out`, and `image_finished`.

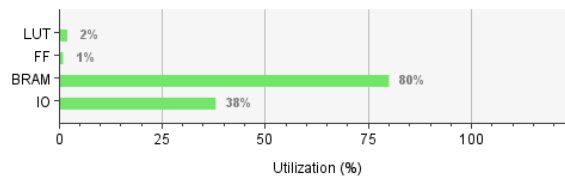
3 FPGA Resource Utilization Comparison: Runtime Reconfigurable vs. Static Version

The following section presents a comparison of FPGA resource utilization between the runtime reconfigurable version (current exercise) and the static version (5th exercise, using generics) for different image sizes, after the Place & Route stage.

3.1 Static Version (Exercise 5)

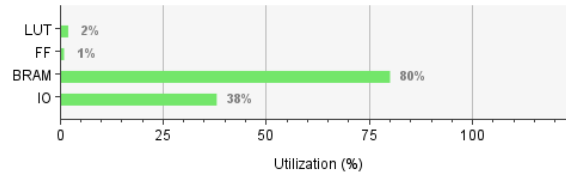
- $N = 256$

Resource	Utilization	Available	Utilization %
LUT	417	17600	2.37
FF	379	35200	1.08
BRAM	48	60	80.00
IO	38	100	38.00



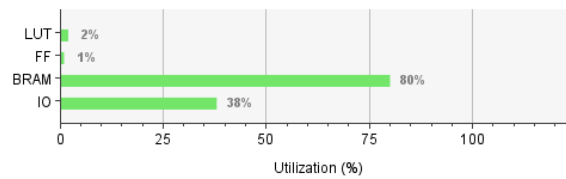
- $N = 512$

Resource	Utilization	Available	Utilization %
LUT	417	17600	2.37
FF	379	35200	1.08
BRAM	48	60	80.00
IO	38	100	38.00



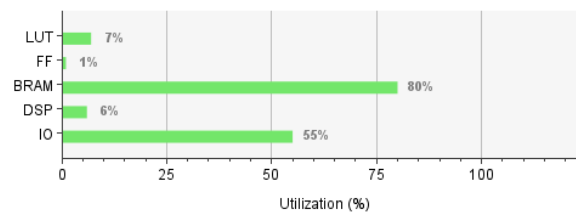
- $N = 1024$

Resource	Utilization	Available	Utilization %
LUT	417	17600	2.37
FF	379	35200	1.08
BRAM	48	60	80.00
IO	38	100	38.00



3.2 Runtime Reconfigurable Version (Exercise 6)

Resource	Utilization	Available	Utilization %
LUT	1190	17600	6.76
FF	395	35200	1.12
BRAM	48	60	80.00
DSP	5	80	6.25
IO	55	100	55.00



3.3 Observations

We observe that the *runtime reconfigurable* version exhibits constant resource utilization, regardless of the size of the image being processed. This occurs because the hardware itself remains unchanged. The relative increase in resource utilization compared to Exercise 5 is due to the addition of communication through `AXI_Lite`, as well as the minor architectural modification described previously, which enables the image size to be changed at runtime.

It should also be noted that the resource utilization is identical for all image sizes in Exercise 5 as well, since a fixed-size FIFO was used in that implementation.

The key advantage of the runtime reconfigurable version is that a single bitstream can support all image sizes. This makes the approach particularly suitable for real-world applications in which the image dimensions may change dynamically.

4 CPU vs. FPGA Speedup Comparison

We measured the execution time of the debayering filter both in software (on the ARM processor) and on the FPGA, for all supported image sizes. The measurements were performed on the Zybo board, with the FPGA operating at a clock frequency of 100 MHz.

Image Size	CPU Cycles	FPGA Cycles	Speedup
256×256	2300313	341895	6.73
512×512	9205303	1367866	6.73
1024×1024	36802087	5453612	6.74

It is worth noting that for images with smaller dimensions, the observed speedup was significantly lower, while for very small images it became almost negligible. We attribute this behavior to a combination of two factors. First, small images are likely to fit entirely within cache memories located close to the ARM core, thereby significantly reducing the execution time of the software implementation. At the same time, the additional overhead introduced by communication through the AXI buses and DMA transfers remains relatively constant regardless of the image size. Therefore, the combination of these two factors may make the use of the FPGA accelerator less advantageous for small images.

5 Simulation Waveform

The waveform of the runtime reconfigurable debayering filter simulation is presented below, verifying the correct operation of the circuit. The configuration phase can be observed first, during which `image_dim_vld` is set to '1' and the image size is configured. This is followed by the pixel reception phase (`valid_in`) and, finally, by the generation of valid output data (`valid_out`).

